Question ID: 568aaf27

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$x+y=12$

$y=x^2$

If (x, y) is a solution to the system of equations above, which of the following is a possible value of x?

Answer

A. 0

B. 1

C. 2

D. 3

Correct Answer: D

Rationale

Choice D is correct. Substituting x^2 from the second equation for y in the first equation yields $x+x^2=12$. Subtracting 12 from both sides of this equation and rewriting the equation results in $x^2+x-12=0$. Factoring the left-hand side of this equation yields $(x-3)(x+4)=0$. Using the zero product property to solve for x, it follows that $x-3=0$ and $x+4=0$. Solving each equation for x yields $x=3$ and $x=-4$, respectively. Thus, two possible values of x are 3 and -4. Of the choices given, 3 is the only possible value of x.

Choices A, B, and C are incorrect. Substituting 0 for x in the first equation gives $0+y=12$, or $y=12$; then, substituting 12 for y and 0 for x in the second equation gives $12=0^2$, or $12=0$, which is false. Similarly, substituting 1 or 2 for x in the first equation yields $y=11$ or $y=10$, respectively; then, substituting 11 or 10 for y in the second equation yields a false statement.